

Chapter One

Introduction to Database Systems

Outline

- Introduction to database system
 - Basic terminologies
- Data handling approaches
- Database & DBMS
- Components of Database System Environment
- Users of Database System

Data VS. Information

- ***Data***
 - is a collection of raw(unorganized) facts.
 - by themselves are useless
- **Information**
 - is a processed data in the form that is meaningful to the user.

Information System

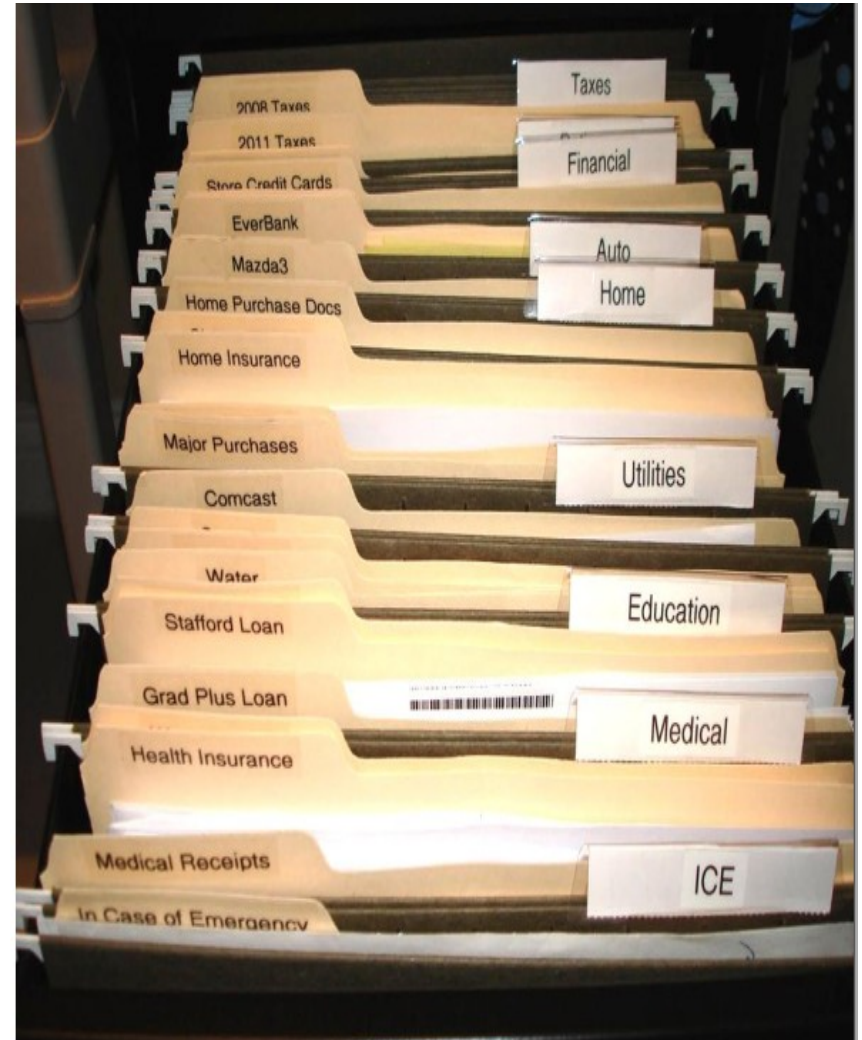
- ***Information System*** is a system that:
 - Receives data and instruction
 - Processes the data as per the instruction
 - Produces output
 - Stores data/information for future use

Data Handling Methods

- There are three data handling methods
 - Manual Approach
 - Computerized Approaches
 - File Based Approach
 - Database Approach

Data handling approaches

- **Manual Approach**
 - data storage and retrieval follows the primitive and traditional way of data/information handling where cards and paper are used for the purpose of keeping records.
 - Typing the data on paper and put in a file cabinet
 - storage and retrieval will be performed using human labor.
 - Works well if the number of items to be stored is small



Limitations of the Manual approach

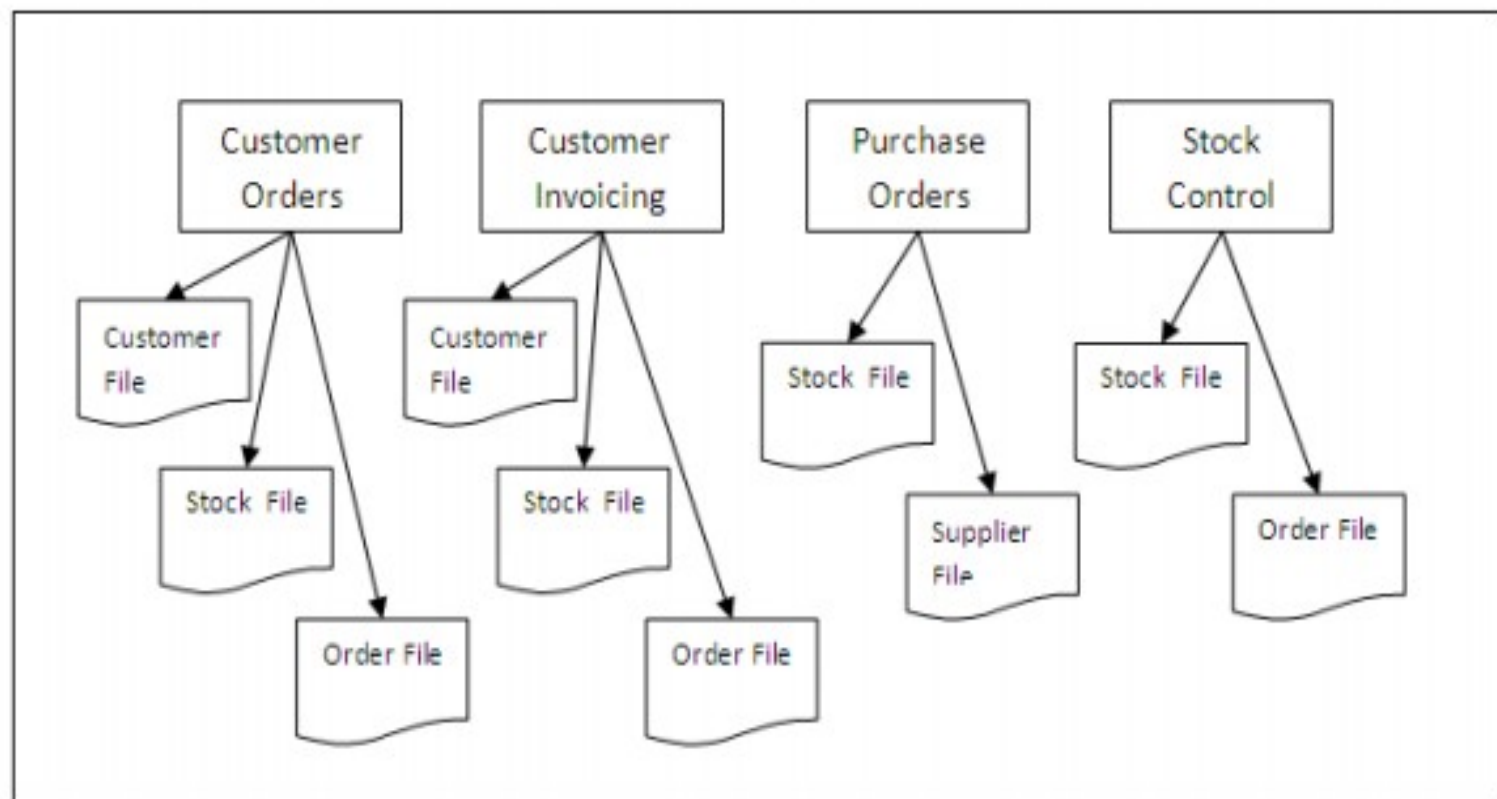
- Prone to **error**
- **Data loss**: due to damaged papers or unable to locate it.
- **Redundancy**: multiple copies of the same data within the organization.
- **Inconsistency**: Modifications are not reflected on all multiple copies.
- Difficult to **update**, retrieve, integrate.
- You have the data but it is difficult to compile the information
- Cross referencing is difficult Cross referencing is difficult . You have the data but it is difficult to compile the information.
- Limited to small size information.

Computerized Approach

- A computerized way of dealing with information
 - File based approach → Decentralized
 - Database approach → Centralized
- Based on where the data resides in the system

File based Approach

- an early attempt to computerize the manual filing system.
- There were, and still are, several computer applications with file based processing for the purpose of data handling.
- In such systems, every application program that provides service to end users **define** and **manage** its own data.
- Such systems have number of programs for each of the different applications in the organization. And this approach is the **decentralized** computerized data handling method.
- File, in file based approach, is a collection of records which contains logically related data .





- A. Students Information
- B. Dorm Information
- C. Books Information
- D. Meals Information
- E. Courses Information

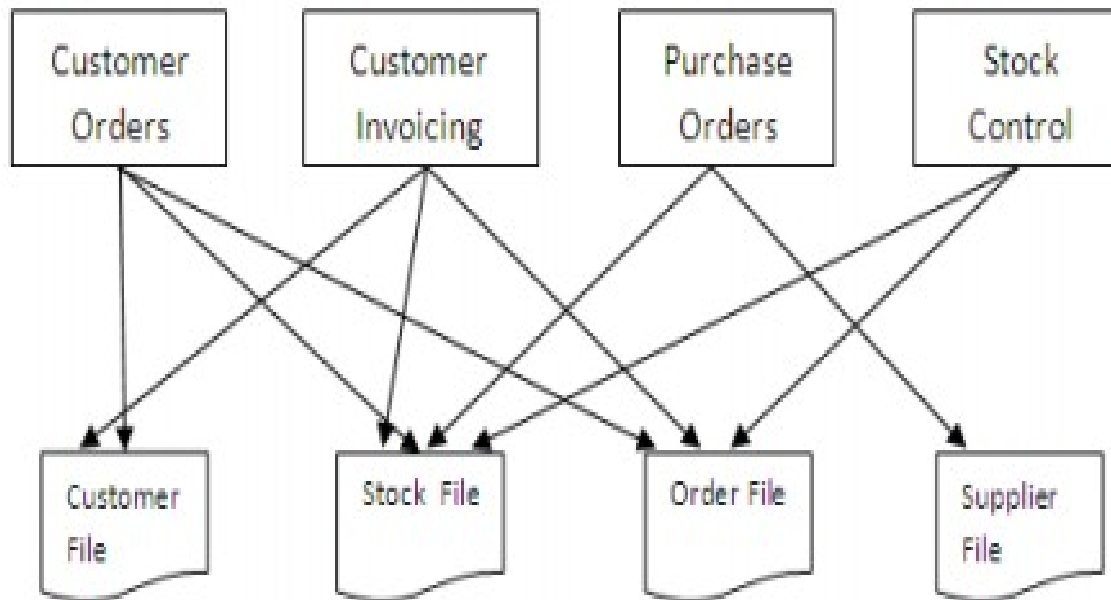
Limitations of the File Based approach

- **Limited data sharing**
- **Duplication of data**
 - Same data is held by different programs
 - Wasted space and potentially different values and/or different formats for the same item
- **Data dependence on the application**
 - Definition of data was embedded in application programs, rather than being stored separately and independently
- **Poor Security and administration**

- **Data Inconsistency and Confusion- Update Anomalies**
 - **Modification Anomalies:** a problem experienced **when one or more data value is modified on one application program but not on others** containing the same data set
 - **Deletion Anomalies:** a problem encountered where **one record set is deleted from one application but remain untouched in other application** programs
 - **Insertion Anomalies:** a problem experienced **when ever there is new data item to be recorded, and the recording is not made in all the applications**

shared file approach

- approach to solving the problem of each application having its own set of files is to **share files** between different applications.



- The introduction of shared files solves the problem of duplication and inconsistent data across different versions of the same file held by different departments, but other problems may emerge, including:
 - File incompatibility
 - Difficult to control access
 - Physical data dependence
 - Difficult to implement concurrency

Database Approach

- Based on
 - The database and
 - Database Management System (DBMS)

Database Approach Cont'd

What is a Database?

- A database is a collection of related data in an organized way.
- of the time, organization is in tabular form.

E.g. book database

<u>Call no</u>	<u>Title</u>	<u>Author</u>	<u>Publisher</u>	<u>No of copies</u>
QA46	Introduction to dbase	Colony	Addison Wesley	15

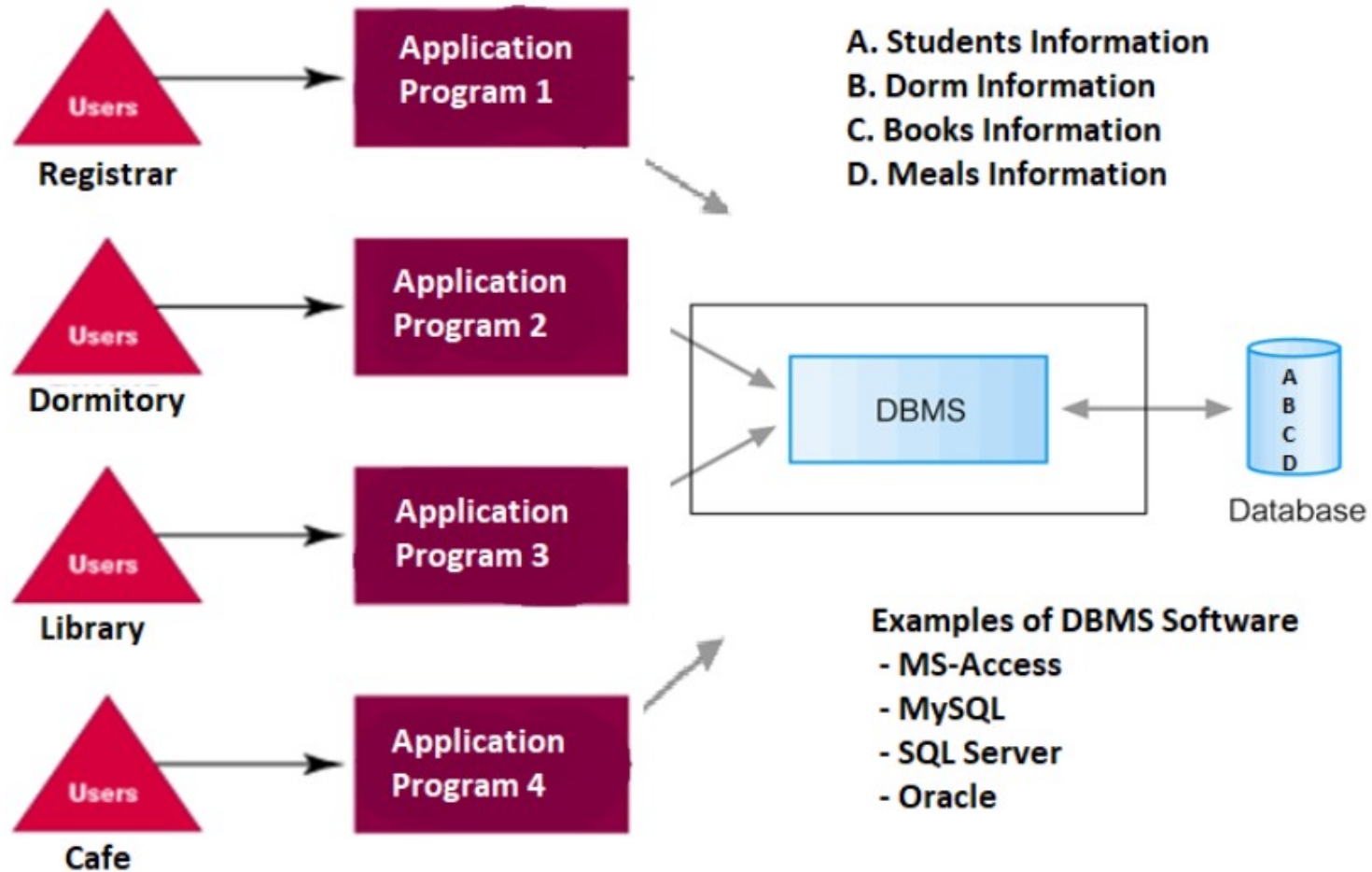
Database Approach Cont'd

- Database
 - is a single repository of data that is defined once and then is accessed by various users.
 - is a shared collection of logically related data and its metadata.
 - Shared collection (Centralized System) – can be used simultaneously by many departments and users.
 - Logically related data comprises entities, attributes, and relationships of an organization's information.
 - In addition to containing data required by an organization, database also contains a description of the data which also called as **“Metadata”** or **“Data Dictionary”** or **“Systems Catalogue”** or **“Data about Data”**

Database Approach Cont'd

- **Database Management System**
 - refers to a software system used to create and manage databases.
 - The term database system is usually an alternative term for database management system.
 - is a general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications
 - **Defining a database** involves specifying the data types, structures, and constraints for the data to be stored in the database
 - **Constructing the database** is the process of storing the data itself on some storage medium
 - **Manipulating a database** includes such functions as querying the database to retrieve specific data, updating the database, and generating reports from the data
 - **Sharing a database** allows multiple users and programs to access the database concurrently

Database Approach Cont'd



Database Approach Cont'd

- DBMS provides the following facilities:
 - **Data Definition Language (DDL)** - Language used to setup a database, create, delete and alter table with the facility of handling constraints
 - **Data Manipulation Language (DML)** - Is a core command used by end users and programmers to store, retrieve, and update the data in the database
e.g. SQL
 - **Data Control Language** - are commands that will help the Database Administrator to control the database. The commands include grant or revoke privileges to access the database.
 - **Data Dictionary** - is used to store and organize information about the data stored in the database

Database Approach Cont'd

DBMS services

1. Data ***storage, retrieval*** and ***update*** in the database.
2. A user accessible ***catalogue***
3. ***Transaction support service***: ALL or NONE transaction, which minimize data inconsistency.
4. ***Concurrency Control Services***: access and update on the database by different users simultaneously should be implemented correctly.
5. ***Recovery Services***: a mechanism for recovering the database after a failure must be available.

Database Approach Cont'd

6. **Authorization**

Services (Security): must support the implementation of access and authorization service to database administrator and users.

7. **Support for Data Communication** : should provide the facility to integrate with data transfer software or data communication managers.

8. **Integrity Services**: rules about data and the change that took place on the data, correctness and consistency of stored data, and quality of data based on business constraints.

9. Services to promote **data independency** between the data and the application

Database Approach Cont'd

Characteristics of the Database Approach

The main characteristics of the database approach are the following:

- 1) Self-describing nature of a database system
- 2) Insulation between programs and data, and data abstraction
- 3) Support of multiple views of the data
- 4) Sharing of data and multiuser transaction processing

1) Self-Describing Nature of a Database System

- Database system contains not only the database itself but also a **complete definition** or **description** of the database structure and constraints.
- The definition contains information like
 - the structure of each file, the type and storage
 - format of each data item, and
 - various constraints on the data.
- These information are called **meta-data** and stored in the catalog.

Cont'd

- The **catalog** is used by the DBMS software and also by database users who need information about the database structure.
- A general-purpose DBMS software package is not written for a specific database application.
 - Therefore, it must refer to the catalog to know the structure of the files in a specific database, such as the type and format of data it will access.

- The DBMS software must work equally well with *any number of database applications—for example, a university database, a banking database, or a company database*—as long as the database definition is stored in the catalog.

- In traditional file processing, data definition is typically part of the application programs themselves. Hence, these programs are constrained to work with only *one specific database*, whose structure is declared in the application programs.

2) Insulation between Programs and Data, and Data Abstraction

- In DBMS the structure of data files is stored in the DBMS catalog separately from the access programs.
 - We call this property **program-data independency**

Example

- Assume we have the following catalog for registrar database.

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

- Internal storage format for a STUDENT record, based on the above database catalog

Data Item Name	Starting Position in Record	Length in Characters (bytes)
Name	1	30
Student_number	31	4
Class	35	1
Major	36	4

- For example, a file access program may be written in such a way that it can access only STUDENT records of the structure shown in the figure above. If we want to add another piece of data to each STUDENT record, say the Birthdate, such a program will no longer work and must be changed. By contrast, in a DBMS environment, we just need to change the description of STUDENT records in the catalog to reflect the inclusion of the new data item Birthdate; no programs are changed. The next time a DBMS program refers to the catalog, the new structure of STUDENT records will be accessed and used.

- In object-oriented and object-relational databases , users can define **operations** on data as part of the database definitions. An operation (also called a function) is specified in two parts.
 - The interface (or signature) of an operation includes the operation name and the data types of its arguments (or parameters).
 - The implementation (or method) of the operation is specified separately and can be changed without affecting the interface.
- User application programs can operate on the data by invoking these operations through their names and arguments, regardless of how the operations are implemented. This may be termed program-operation independence.
- The characteristic that allows program-data independence and program-operation independence is called **data abstraction**.

- A DBMS provides users with a **conceptual** representation of data.
 - This does not include many of the details of how the data is stored or how the operations are implemented.

- Database users and application programs refer to the conceptual representation of the files, and the DBMS extracts the details of file storage from the catalog when these are needed by the DBMS file access modules.

3) Support of Multiple Views of the Data

- Each users of a database may require a different perspective or view
- A view may be a subset of the database or it may contain **virtual data** that is derived from the database files but is **not explicitly stored**.
- A multiuser DBMS whose users have a variety of distinct applications must provide facilities for defining multiple views.

Example

Assume we have data about

- course
- section
- grade_report
- prerequisite

- One user of the database may be interested only in accessing and printing the transcript of each student

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

- A second user, who is interested only in checking that students who have taken all the prerequisites of each course for which they register

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

4) Sharing of Data and Multiuser Transaction Processing

- A transaction is an *executing program or process that includes one or more database accesses*, such as reading or updating of database records.
- A multiuser DBMS must allow multiple users to access the database at the same time.

- The DBMS must include **concurrency** control software to ensure that several users trying to update the same data do so in a controlled manner so that the result of the updates is correct.

Database Management System(DBMS)

- A full scale DBMS should at least have the following services to provide to the user
 - **Data storage, retrieval and update** in the database
 - A user accessible **catalogue**
 - **Transaction support service**: ALL or NONE transaction, which minimize data inconsistency
 - **Concurrency Control Services**: access and update on the database by different users simultaneously should be implemented correctly
 - **Recovery Services**: a mechanism for recovering the database after a failure must be available
 - **Authorization Services (Security)**: must support the implementation of access and authorization service to database administrator and users
 - **Support for Data Communication**: should provide the facility to integrate with data transfer software or data communication managers
 - **Integrity Services**: rules about data and the change that took place on the data, correctness and consistency of stored data, and quality of data based on business constraints

- organization of the database becomes necessary when the data is voluminous. Otherwise, managing data will be very difficult.

E.g.

- A Manufacturing Company with product data
- A Bank with account data
- A Hospital with patients
- A University with Students
- A government with planning data

Advantages of a database approach

- Ease of application development
- Minimal data redundancy
- Physical data dependence is resolved
- Compactness
- Speed
- Accuracy
- Data can be shared
- Inconsistency can (to some extent) avoided
- Standards can be enforced
- Security restrictions can be applied
- Integrity can be maintained
- Conflicting requirements can be balanced
- Transaction support can be provided
- Improved decision support
- Less labor

- **Limitations and risk of Database Approach**

- Introduction of new professional and specialized personnel.
- Complexity in designing and managing data
- The cost and risk during conversion from the old to the new system
- High cost to be incurred to develop and maintain the system
- Complex backup and recovery services from the users perspective
- Reduced performance due to centralization and data independency
- High impact on the system when failure occurs to the central system.

Components of a Database System

- four major components,
 - data
 - hardware
 - software
 - users and designers of database

- **Data:**
- **Issues**
- stored as a single database or distributed
- Is the system a single-user or multi-user one?
- How are we going to achieve the utmost possible performance concerning the data storage and maintenance?
- What other benefits or drawbacks do we expect as the result of placement or structure of the database?

- **Hardware:**
- secondary storage media (disks, tapes and optical media)
- controllers (hard disk controller, etc.); and
- the processor(s) and associated main memory

- **Software:** Database Management System (DBMS)
 - Is responsible for the overall management of communications between the user and the database.
 - It is found between the data and the users, which, in other words, means the data is entirely covered or shielded by the DBMS software.
 - The DBMS provides facilities for operating on the database.

Actors on the Scene

- People whose jobs involve the day-to-day use of a large database; we call them the *actors on the scene*.
- Peoples whose work is to maintain the database system environment but who are not actively interested in the database contents---are called *workers behind the scene*

Cont'd

- **Actors On the Scene:**
 - Database Administrator
 - Data Administrator
 - Database Designer
 - End Users

Database Administrator (DBA)

- Responsible to oversee, control and manage the database resources (the database itself, the DBMS and other related software)
- Authorizing access to the database
- Coordinating and monitoring the use of the database
- Responsible for determining and acquiring hardware and software resources
- Accountable for problems like poor security, poor performance of the system
- Involves in all steps of database development
- We can have further classifications of this role in big organizations having huge amount of data and user requirement.

- **Data Administrator (DA):** is responsible on management of data resources. Involves in database planning, development, maintenance of standards policies and procedures at the conceptual and logical design phases.

Database Designer (DBD)

- Identifies the data to be stored and choose the appropriate structures to represent and store the data.
- Should understand the user requirement and should choose how the user views the database.
- Involve on the design phase before the implementation of the database system.

We have two distinctions of database designers, one involving in the logical and conceptual design and another involving in physical design.

Logical and Conceptual DBD

- Identifies data (entity, attributes and relationship) relevant to the organization
- Identifies constraints on each data
- Understand data and business rules in the organization
- Sees the database independent of any data model at conceptual level and consider one specific data model at logical design phase.

Physical DBD

- Take logical design specification as input and decide how it should be physically realized.
- Map the logical data model on the specified DBMS with respect to tables and integrity constraints. (DBMS dependent designing)
- Select specific storage structure and access path to the database
- Design security measures required on the database

Application Programmer and Systems Analyst

- **System analyst** determines the user requirement and how the user wants to view the database.
- The **application programmer**
 - who are responsible for writing application programs that use the database using some programming language such as COBOL, Pascal, or a programming language built-in to the DBMS.
 - implements these specifications as programs; code, test, debug, document and maintain the application program.
 - Determines the interface on how to retrieve, insert, update and delete data in the database.
 - The application could use any high level programming language according to the availability, the facility and the required service.

End-users

These are those people who are engaged on processing different types of operations on the database system. Users are workers, whose job requires accessing the database frequently for various purpose. There are different group of users in this category.

Naïve Users:

- Sizable proportion of users
- Unaware of the DBMS
- Only access the database based on their access level and demand
- Use standard and pre-specified types of queries.
- ***Eg. Bank tellers ,reservation agents for airlines, hotels, and car rental companies***

Sophisticated Users

- Are users familiar with the structure of the Database and facilities of the DBMS.
- Have complex requirements
- Have higher level queries
- Are most of the time engineers, scientists, business analysts, etc

Casual Users

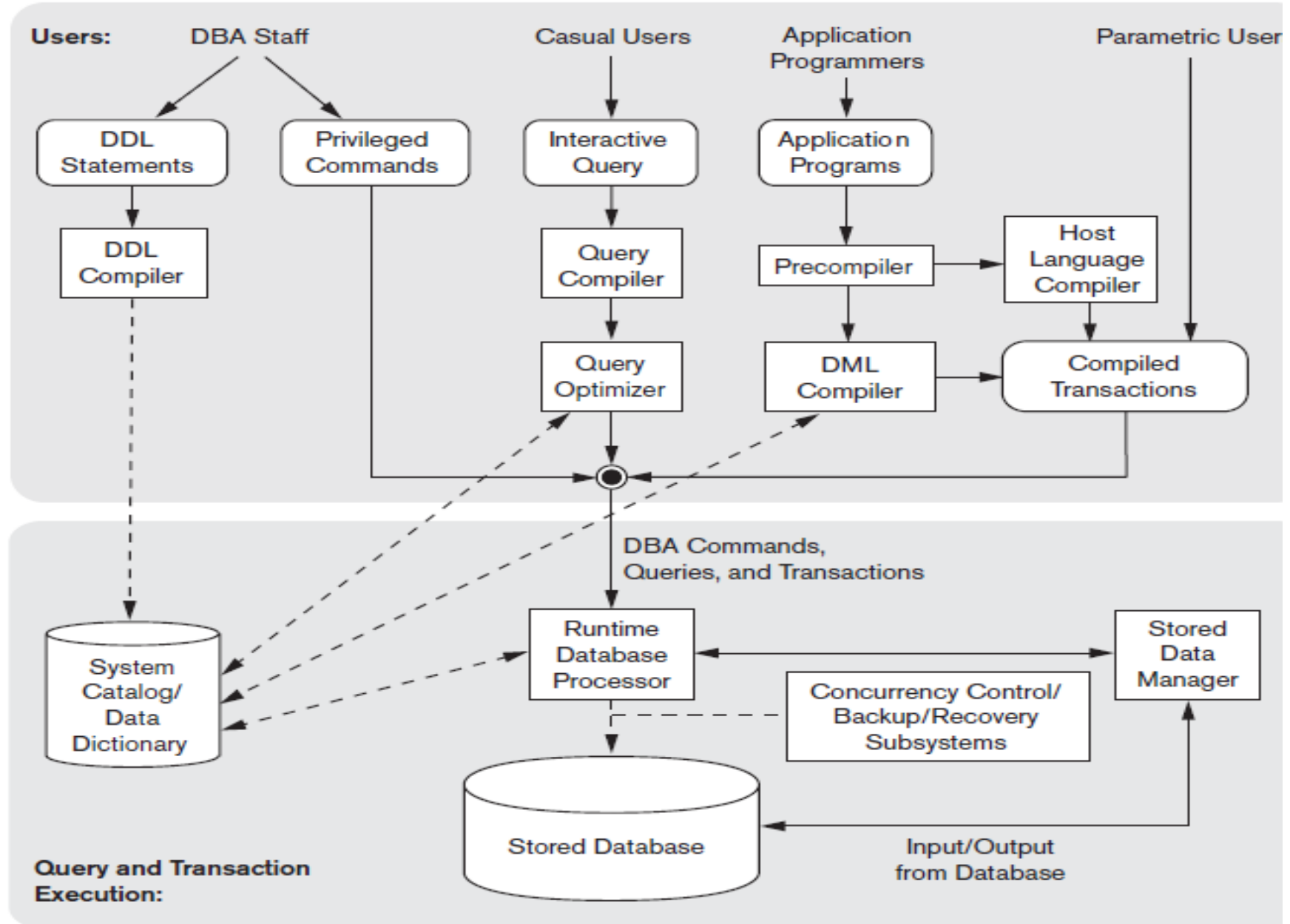
- Users who access the database occasionally.
- Need different information from the database each time.
- Use sophisticated database queries to satisfy their needs.
- Are most of the time middle to high level managers.,

Workers behind the Scene

- DBMS system designers and implementers
 - design and implement the DBMS modules and interfaces as a software package.
- Tool developers
 - design and implement **tools**
 - software packages that facilitate database modeling and design, database system design, and improved performance.

- Operators and maintenance personnel
 - system administrator
 - are responsible for the actual running and maintenance of the hardware and software environment for the database system.

Database System environment



- The database and the DBMS catalog are usually stored on disk.
- Access is controlled by
 - Operating system (OS)
 - buffer management module
 - stored data manager module

- The DDL compiler
 - processes schema definitions, specified in the DDL, and
 - Stores descriptions of the schemas (meta-data) in the DBMS catalog.

- Casual users interact with the database with the **interactive query interface**
- then the query compiler
 - parse and validate the correctness of
 - the query syntax
 - the names of files and data elements
 - then compiles them into an internal form.
 - This internal query is subjected to query optimization

- query optimizer is concerned with the
 - rearrangement and possible reordering of operations,
 - elimination of redundancies, and
 - use of correct algorithms and indexes during execution.
- consults the system catalog
- Generates executable code that performs the necessary operations for the query and makes calls on the runtime processor.

- The **precompiler extracts DML commands** from an application program written in a host programming language.
 - It sends the commands to DML compiler
- The rest of the program is sent to the host language compiler.

- The **runtime database processor executes**
 - ¬the privileged commands,
 - ¬the executable query plans, and
 - ¬the canned transactions with runtime parameters.
- It works with the
 - ¬system catalog
 - ¬stored data manager
- concurrency control and backup and recovery systems are integrated into the working of the runtime database processor for purposes of transaction management.

Database System Utilities

- help the DBA manage the database system.
 - Loading:** A loading utility is used to load existing data files—such as text files or sequential files—into the database.
 - Backup:** A backup utility creates a backup copy of the database, usually by dumping the entire database onto tape or other mass storage medium.

- **Database storage reorganization:** used to reorganize a set of database files into different file organizations, and create new access paths to improve performance.
- **Performance monitoring:** Such a utility monitors database usage and provides statistics to the DBA.

Database Development Life Cycle

The major steps in database system development are;

1. Planning

2. Analysis

3. Design

- **Conceptual Design**

- **Logical Design**

- **Physical Design**

4. Implementation

5. Operation and Support

